

MTO-based Localized Orbital (MLO) — sample tests and how to run

`mlo` builds a localized real-space tight-binding-style Hamiltonian on top of ecalj's PMT (MTO + APW) basis. This directory covers semiconductors, magnetic metals, multilayers, and 4f systems — see the table below for the full list. The SOC-as-perturbation variant (`job_mlo_soc`) is documented separately in [README_SOC.md](#).

Quick start

```
cd ~/ecalj/Samples/MLOsamples
# SOC test set (job_mlo + job_mlo_soc)
testecalj -np 8 GaAsSoc      # semiconductor, with SOC
testecalj -np 8 FeSoc        # ferromagnetic Fe (nspin=2), with SOC
testecalj -np 8 FeMgOSoc     # FeMgO multilayer (nspin=2), with SOC

# Plain MLO non-SOC
testecalj -np 8 Si666gwsc     # Si QSGW, non-magnetic, MLO + DFT band
check
testecalj -np 8 Al2O3_Cr      # Cr-doped Al2O3 QSGW80, mlo_emax=7
example
testecalj -np 8 C             # diamond C
testecalj -np 8 C.sp          # graphite-like C (sp-only)
testecalj -np 8 Cu            # fcc Cu
testecalj -np 8 SrTiO3        # perovskite, non-magnetic
testecalj -np 8 Fe            # bcc Fe (nspin=2, no SOC)
testecalj -np 8 FeCo          # FeCo alloy (nspin=2)
testecalj -np 8 FeMgO         # FeMgO multilayer (nspin=2, no SOC)
testecalj -np 8 GdCo5         # GdCo5 (Gd 4f, nspin=2)
testecalj -np 8 GdION         # Gd ion QSGW
testecalj -np 8 NiO666lda     # NiO LDA AFM
testecalj -np 8 RuO2          # RuO2 QSGW
testecalj -np 8 SmP           # SmP (Sm 4f + so=2)
```

Each test runs the full pipeline (lmf → mlo, plus the 4-step

`job_mlo_soc` flow on `*Soc` samples), then numerically diffs the produced `band_MLO_spin{1,2}*.dat` against committed references with tolerance $7.4\text{e-}5 \text{ Ry} \approx 0.001 \text{ eV}$. PASS line:

```
max deviation = 0.0  tolerance = 7.4e-05
```

Sample summary

dir	system	nspin	SOC test	reference
Si666gwsc	Si QSGW (6×6×6)	1	–	bnd00{1..6}.spin1, band_MLO_spin1.dat
GaAsSoc	GaAs QSGW	1	✓	band_MLO_spin1.dat, band_MLO_spin1.soc.dat
FeSoc	bcc Fe DFT	2	✓	band_MLO_spin{1,2}.dat, band_MLO_spin1.soc.dat
FeMgOSoc	FeMgO multilayer DFT	2	✓	band_MLO_spin{1,2}.dat, band_MLO_spin1.soc.dat
Al2O3_Cr	Cr-doped Al2O3 QSGW80	2	–	band_MLO_spin{1,2}.dat (mlo_emax=7)
C	diamond C	1	–	band_MLO_spin1.dat
C.sp	C graphite-like (sp)	1	–	band_MLO_spin1.dat
Cu	fcc Cu	1	–	band_MLO_spin1.dat
SrTiO3	SrTiO3 perovskite	1	–	band_MLO_spin1.dat
Fe	bcc Fe DFT (no SOC)	2	–	band_MLO_spin{1,2}.dat
FeCo	FeCo alloy	2	–	band_MLO_spin{1,2}.dat
FeMgO	FeMgO multilayer	2	–	band_MLO_spin{1,2}.dat
GdCo5	GdCo5 (4f magnetic)	2	–	band_MLO_spin{1,2}.dat
GdION	Gd ion QSGW	2	–	band_MLO_spin{1,2}.dat
NiO666lda	NiO LDA AFM	2	–	band_MLO_spin{1,2}.dat
RuO2	RuO2 QSGW	2	–	band_MLO_spin{1,2}.dat
SmP	SmP (4f, so=2 Lz-Sz)	2	–	band_MLO_spin{1,2}.dat

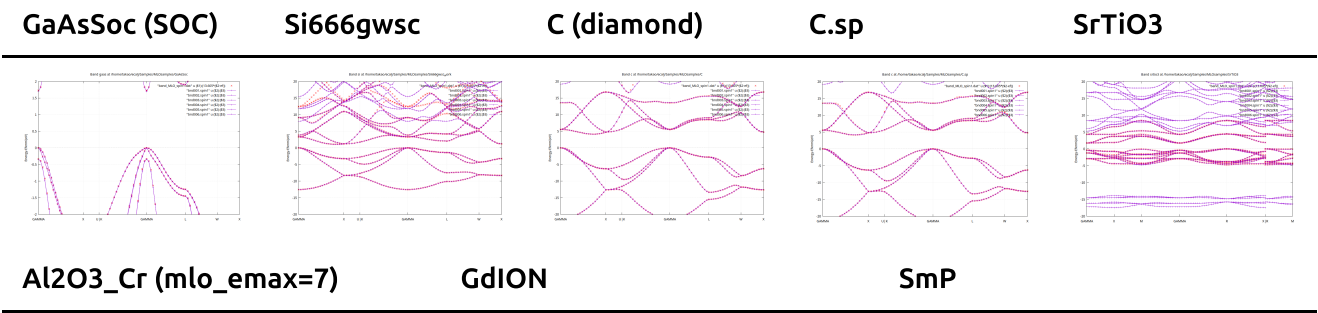
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For `Si666gwsc` the MLO test was added on top of an existing `job_band` DFT-band check, so `testecalj` runs both. `Al2O3_Cr` is the canonical example for `mlo_emax > 0` (Cr 3d states sit ~7 eV above E_F).

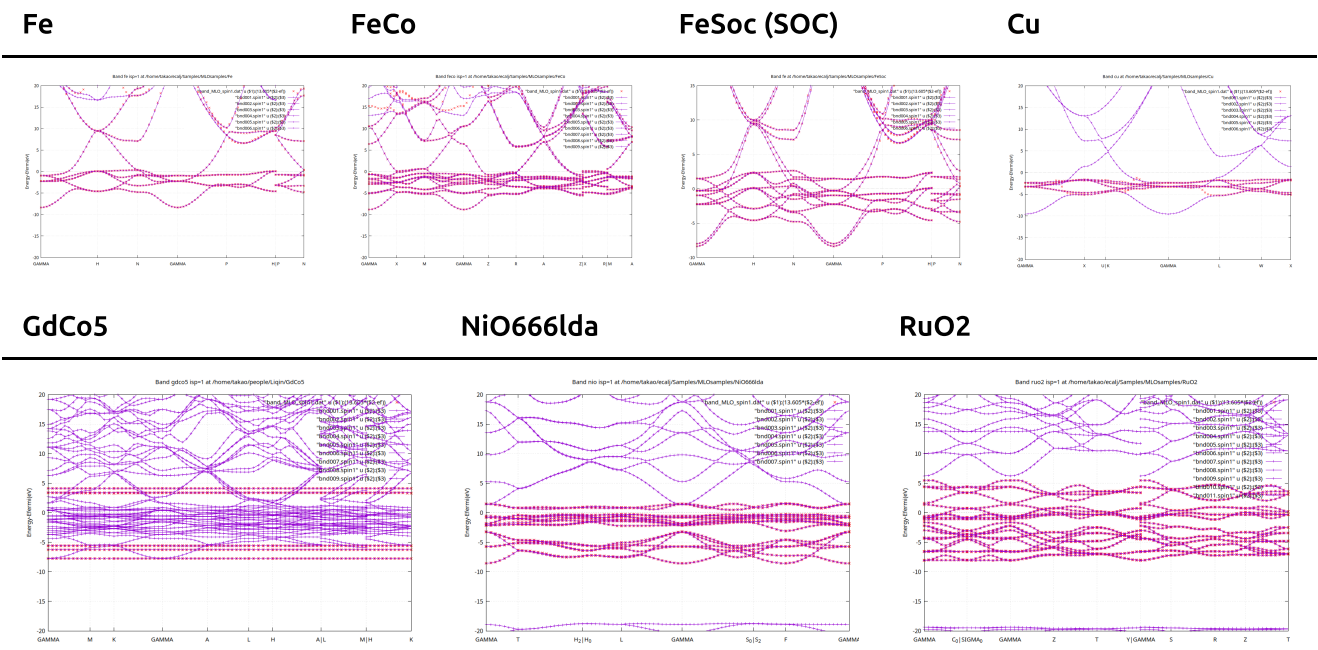
Plots

All plots are `bandplot_MLO.isp1.glt` rendered after `testecalj` finishes. Red dots = MLO bands, lines = full DFT bands. SOC samples show the 2N-spinor MLO overlay (post-`job_mlo_soc`).

Semiconductors / insulators



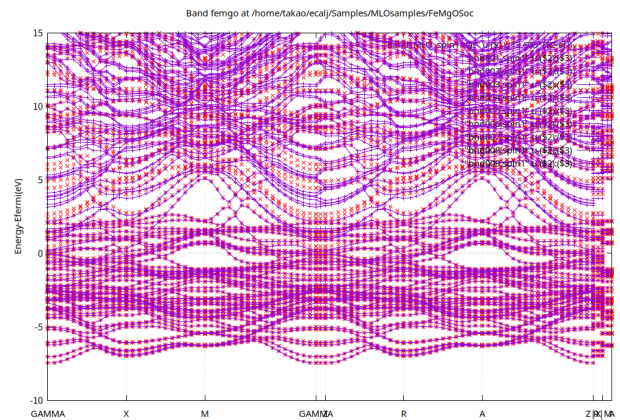
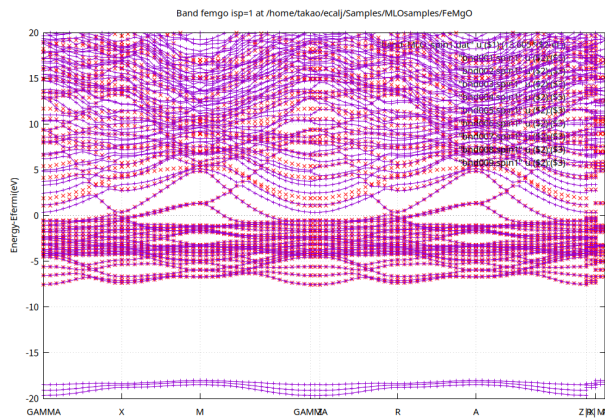
Magnetic metals



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Multilayers





Manual run

```
cd <target>_work          # produced by testecalj
gnuplot -p bandplot_MLO.isp1.glt      # red = MLO bands, line = DFT
bands
gnuplot -p bandplot_MLO.isp2.glt      # spin-down (FeSoc / FeMgOSoc
only)
```

For SOC samples, the same `bandplot_MLO.isp1.glt` shows the 2N-spinor MLO overlay after `job_mlo_soc` finishes (Fermi level reset to `efermi_soc`). See `README_SOC.md` for what to expect.

To run only the MLO step manually:

```
job_mlo gaas -np 8          # non-SOC
job_mlo_soc gaas -np 8      # SOC as perturbation
```

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What gets shown in the plot

Both DFT and MLO bands plotted on the same panel (Energy - E_F , eV).

- `Si666gwsc`: dispersive sp bands. MLO traces the lower 4 valence + a few conduction bands faithfully along Γ -X-U- Γ -L-W-X.
- `GaAsSoc`: VBM around Γ shows the As-p triplet. With `job_mlo_soc`, the $\Delta_{SO} \approx 0.34$ eV split-off band is reproduced.
- `FeSoc`: Fe 3d manifold ± 2 eV around E_F , both spins.
- `FeMgOSoc`: dense band structure of the Fe-MgO interface region.

Settings to know (in `ctrlG.<sname>.toml`)

As of 2026-05 the Fortran binaries read structured TOML only. Legacy `GWinput` is converted on first use by `Legacy2toml.py`; what follows

shows the converted form actually consumed by `lmf/mlo`.

```
mlo_method = 0      # 0 = standard, 1 = emax-cut, 2 = MTO-energy cut
mlo_emax    = 0      # eV relative to E_F (semiconductor: 0 ≈ at E_F).
                  # User-set 0 is honoured literally; sentinel

default for
                  # "key absent" applies emax*Ry at runtime.
                  # Al2O3_Cr-style insulators: try 7 eV.

Worb = ""          # which lm orbitals per atom enter the MLO basis
  1 Ga   1 2 3 4 5 6 7 8 9
  2 As   1 2 3 4 5 6 7 8 9
""
```

The `Worb` indices map to (l, m) : $1 \rightarrow s$; $2..4 \rightarrow p$; $5..9 \rightarrow d$; $10..16 \rightarrow f$
(see `bandplot.isp1.glt` header for the full table).

Screened W on the MLO basis (separate flow)

```
job_mloW fe -np 8
```

Computes the on-site interaction strength on the MLO basis. The two output blocks are written separately so users can post-process them:

- `Coulomb_v.{UP,DN}` — bare V on MLO basis
- `Screening_W-v.{UP,DN}` — RPA-screened $W - V$ on MLO basis
- The full screened W is then **the sum** of the two.

What actually matters for downstream analysis:

- **On-site (one-atom) blocks** of V and $W - V$ — the U/J parameters in Hubbard-style models live here.
- **W at $\omega = 0$** (static screened interaction).

The full ω -dependence is stored on disk (frequency mesh from `HistBin_dw/HistBin_ratio`), but for typical use cases only the $\omega = 0$ slice and the on-site sub-block need post-processing.

Reference value: bcc Fe, Fe-3d on-site W

Indicative ballpark (Takao's 2026-02-04 follow-up after the initial working commit `6e4d7500a`): the on-site screened W on the Fe 3d block comes out around **~1.5 eV**.

NOTE: We need `--mlo_diagnormalization` so that $\delta_{1=|i|}^i = 1$.

Reproduction runs (2026-05-08, gfortran-14, ROMIO)

Re-running `job_mloW fe -np <N>` in `Samples/MLOsamples/Fe/` with `mlo_method = 0, --mlo_diagnorm` (per current `job_mloW` script), on-site Fe-1 d-block diagonal `<i i | W | i i>` at $R=0$, $\omega=0$:

BZ mesh	parallelism	W (t2g)	W (eg)
2×2×2	-np 1	2.27 eV	2.38 eV
2×2×2	-np 4	2.27 eV	2.38 eV (bit-identical to -np 1)
4×4×4	-np 4	1.69 eV	1.81 eV

Trend: denser BZ mesh → smaller W (more screening channels), converging toward Takao's ~1.5 eV ballpark.

This is plain RPA, not cRPA: the cRPA block in `job_mloW` is commented out (a minor follow-up is needed to enable it cleanly).

A live reference run lives in `Fe_job_mloW/` — bcc Fe with `Coulomb_v.{UP,DN}` and `Screening_W-v.{UP,DN}` on the Fe-3d MLO basis.

MPI-IO note: force ROMIO, not OMPIO

`job_mloW` sets `export OMPI_MCA_io=romio321` near the top.

OpenMPI 4.x ships two MPI-IO implementations. The default OMPIO uses a POSIX semaphore in `/dev/shm/sem.OMPIO_<filename>` for file locking. If a previous `hx0fp0` was killed (SIGKILL or out-of-memory), that semaphore is **not** cleaned up; the next `mpi_file_open(__WVI.<iq>)` on the same file name then waits on it forever and the pipeline hangs in `futex_do_wait`. ROMIO uses kernel `fcntl` advisory locks, which are auto-released on process death, so stale state cannot accumulate. The switch is local to `job_mloW`'s environment, so concurrent users or other GW jobs on the same machine are not affected.

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Symptom (if you see it elsewhere): `hx0fp0` running with 0 % CPU and `/proc/<pid>/maps` shows a mapped `/dev/shm/sem.OMPIO_<file>`. Recovery is `rm -f /dev/shm/sem.OMPIO_*` **only when no other MPI-IO job is running on the box** — on a shared machine it is safer to leave the job to ROMIO via the env var rather than rely on global cleanup.

Recent fixes (2026-05)

The MLO + TOML migration introduced two regressions, fixed in commits `10e642915 / c11663856 / 86e58d0c4 / 802878e82`:

- `mlo_emax 0`: TOML reader treated 0 as "use default", overriding user intent. Now sentinel-based — explicit 0 is honoured.
- **<Word> duplicate blocks**: `gwinput2toml.py` last-write-wins silently dropped real data when GWinput had a real block plus a commented

example. Now keeps the first.

- **job_mlo_soc -v overrides:** `-vnspin=2 -vso=0` form is silently ignored under TOML-only mode; replaced with `-v[ham.nspin]=2, -v[ham.so]=0/1`.
- **gfortran 13.3 / 14.2 codegen bug** in `m_HamPMT.f90` (`ReadInfoFromGWinput` block): a "naked" `else: call rx` on the `<Worb>` if/else miscompiles the live (TOML) path, corrupting `lindex` and surfacing as `zhev_tk2: nev /=nev x ...` at runtime. Bisect (2026-05-07) found the empirically minimum bait that prevents the miscompile is one line — `aaa = trim(aaa) // ' '` — both `trim()` and `//` are required (either alone fails). The first if (`mlomethod` scalar) is unaffected and needs no bait. The trigger is *not* partial-array assignment as initially suspected. All 17 MLO samples verified with gfortran 14.2.0.

See `README_SOC.md` for SOC-specific implementation notes.